

Central Dogma of Molecular Biology



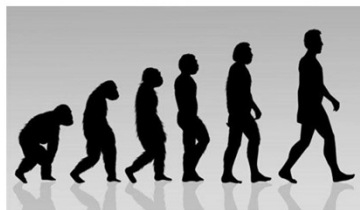
R. K. Gothwal, PhD
Department of Bioengineering and Biotechnology
BIT Mesra, Ranchi
Jaipur-Campus

What is Molecular Biology?

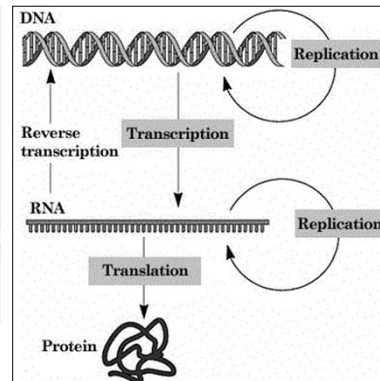
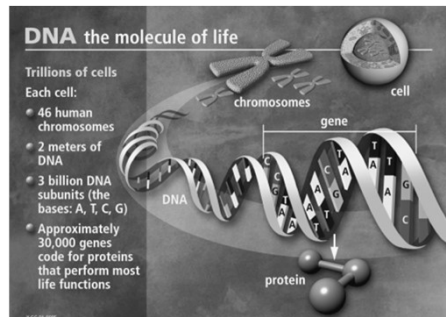
The branch of biology that deals with the nature of biological phenomena at the molecular level through the study of DNA, RNA, proteins, and other macromolecules involved in **genetic information** and **cell function**.



Evolution



Central Dogma of Molecular Biology



Reverse transcription is the transfer of information from RNA to DNA (the **reverse** of normal transcription). This is known to occur in the case of retroviruses, such as

- ❖ **Tobacco Mosaic Virus**
- ❖ **Rous sarcoma**
- ❖ **SARS-CoV-2**

Dogma means “a body of belief”

Module - II

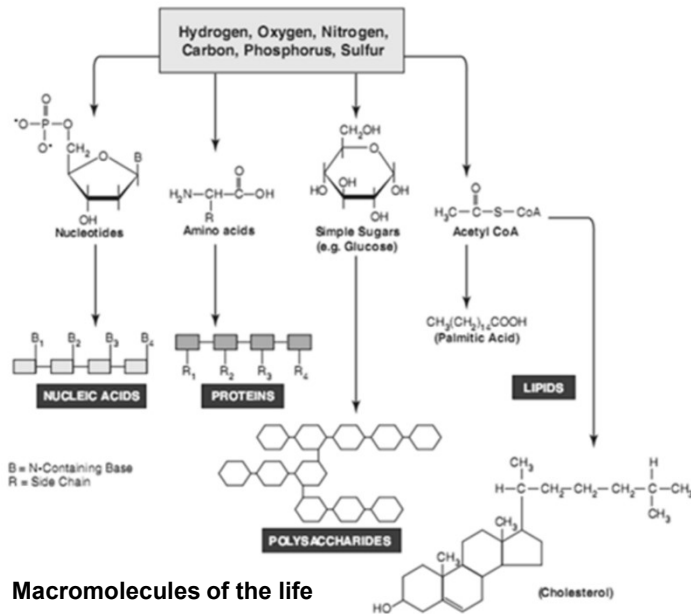
Biomolecules

Nucleic Acids



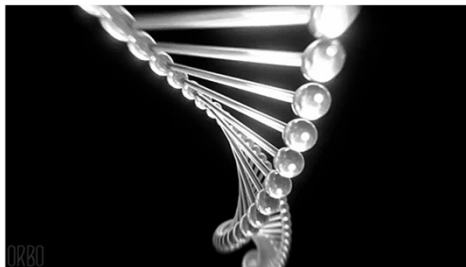
R. K. Gothwal, PhD
Department of Bioengineering and Biotechnology
BIT Mesra, Ranchi
Jaipur-Campus

Biomolecules



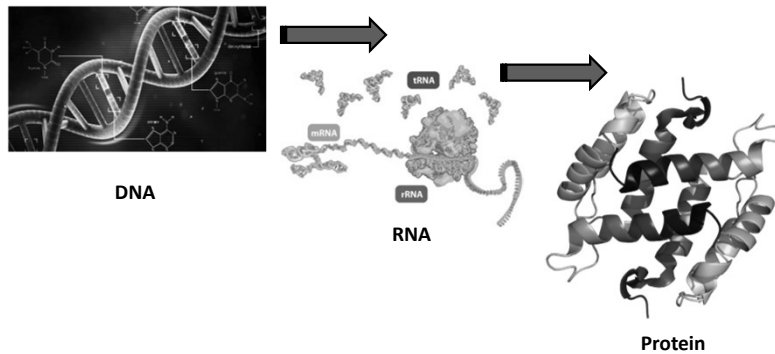
Nucleic Acids

- ❖ Nucleic acids are the most important biological macromolecules which is essential for all living organisms.
- ❖ They function in encoding, transmitting and expressing genetic information.
- ❖ Nucleic acids constitute the genetic material present in all living systems.
- ❖ This genetic material carries the information about the reconstruction of living organism which is in the form of a code, called **Genetic Code**. This code is essential to regulate and propagate life.

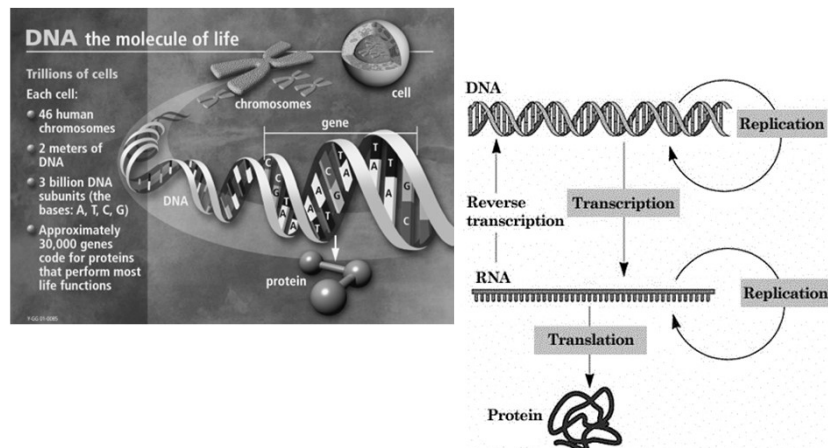


What is Molecular Biology?

The branch of biology that deals with the nature of biological phenomena at the molecular level through the study of DNA and RNA, proteins, and other macromolecules involved in genetic information and cell function.



Central Dogma of Molecular Biology



Francis Crick (1970). *Nature*, 227: 561-563.

Nucleic Acids

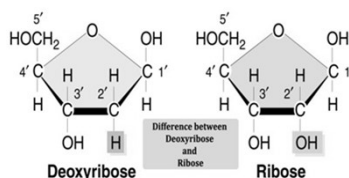
Types

- ❖ Deoxyribose nucleic acid (**DNA**): Double stranded contain deoxy ribose sugar.
- ❖ Ribose nucleic acid (**RNA**) : Single stranded contain ribose sugar.
- ❖ **Size**: Nucleic acids can vary in size, but are generally very large molecules. Nucleic acid molecules range in size from 21 nucleotides (small interfering RNA) to large chromosomes (human chromosome-1 is a single molecule that contains 247 million base pairs).
- ❖ **Biopolymer**: Nucleic acid is the largest biopolymer consists of monomeric units called nucleotides.

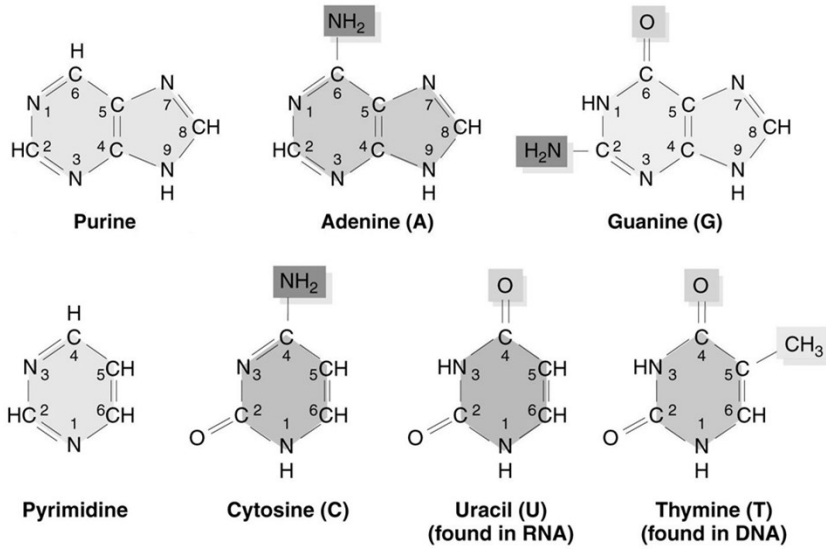
Nucleic Acids

Nucleic acids are long chain molecules formed of several repeating units called Nucleotides. The Nucleotides are composed of 3 important chemical components:

- ❖ **Sugar**: It is a pentose sugar (5 carbon atoms) having a pentagonal ring structure. It is Deoxyribose sugar in DNA while in RNA it is Ribose sugar.
- ❖ **Phosphate**: It is present as Phosphoric acid (H_3PO_4).
- ❖ **Nitrogen Bases**: Based on their chemical structure they are grouped into 2 classes:
 - **Purines** which include Adenine (A) and Guanine (G) and
 - **Pyrimidines** which include Cytosine (C), Thymine (T) and Uracil (U).

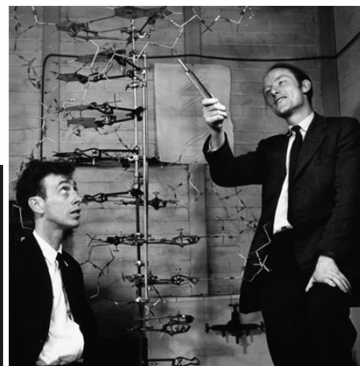


Nitrogen Bases



Structure of DNA

- DNA is a double helical structure
- Made up of two nitrogen bases
 - Purines- Adenine & Guanine
 - Pyrimidines- Cytosine & Thymine

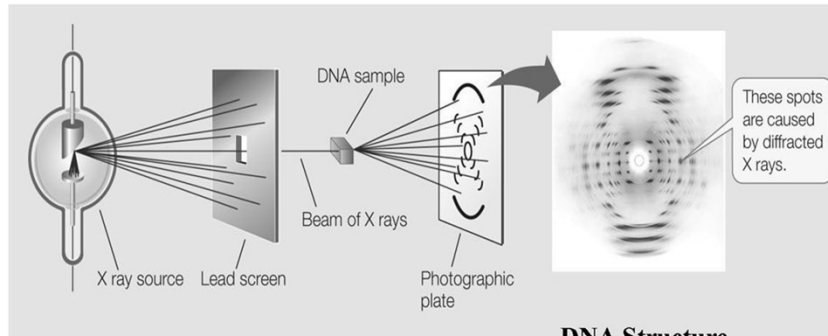


James Watson

Francis Crick

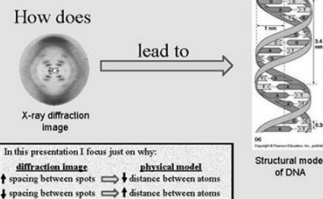
Francis Crick (1970). *Nature*, 227: 561-563.

Structure of DNA

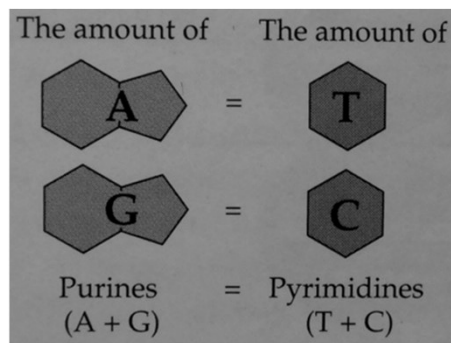


X-Ray Crystallography of DNA

DNA Structure



Chargaff's Rule



Sugar

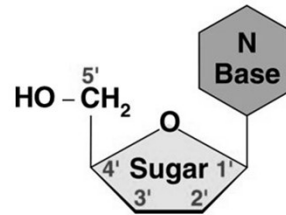
- Ribose - RNA
- deoxyribose - DNA

T	A	C	G	G	C	A	T	A	G	A
A	T	G	C	C	G	T	A	T	C	T

Nucleoside

A nucleoside:

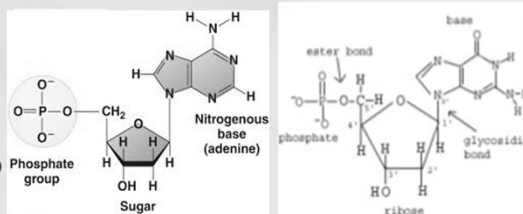
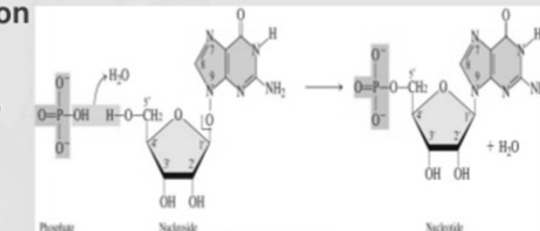
- Has a nitrogen base linked by a glycosidic bond to C1' of a ribose or deoxyribose.
- Is named by changing the the nitrogen base ending to *-osine* for purines and *-idine* for pyrimidines



Nucleotide Formation

Nucleotide Formation

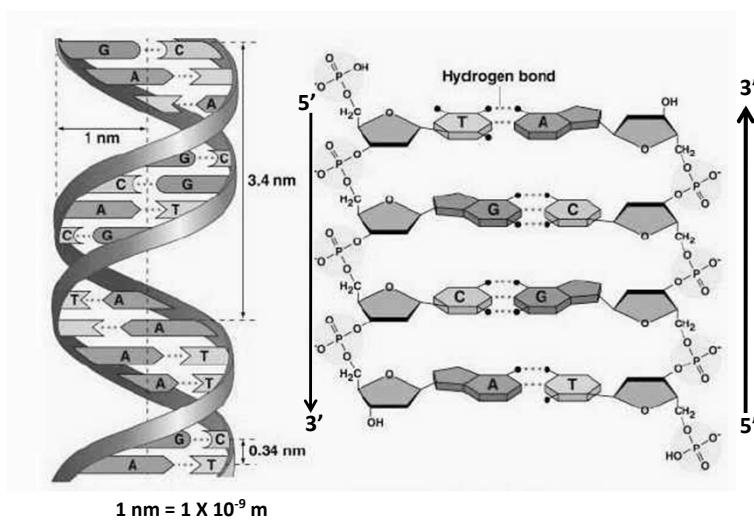
- Phosphate group is added to a nucleoside
 - attached to C_{5'} position through a phosphoester bond
 - condensation reaction (H₂O released)
 - named by appending 5' - monophosphate to nucleoside name



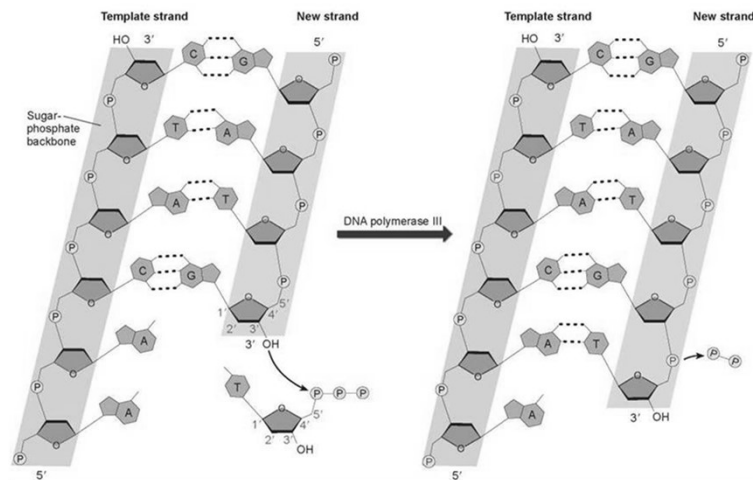
Names of Nucleosides and Nucleotides

Base	Nucleosides	Nucleotides
RNA		
Adenine (A)	Adenosine (A)	Adenosine 5'-monophosphate (AMP)
Guanine (G)	Guanosine (G)	Guanosine 5'-monophosphate (GMP)
Cytosine (C)	Cytidine (C)	Cytidine 5'-monophosphate (CMP)
Uracil (U)	Uridine (U)	Uridine 5'-monophosphate (UMP)
DNA		
Adenine (A)	Deoxyadenosine (A)	Deoxyadenosine 5'-monophosphate (dAMP)
Guanine (G)	Deoxyguanosine (G)	Deoxyguanosine 5'-monophosphate (dGMP)
Cytosine (C)	Deoxycytidine (C)	Deoxycytidine 5'-monophosphate (dCMP)
Thymine (T)	Deoxythymidine (T)	Deoxythymidine 5'-monophosphate (dTMP)

Structure of DNA



Structure of DNA

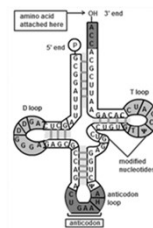
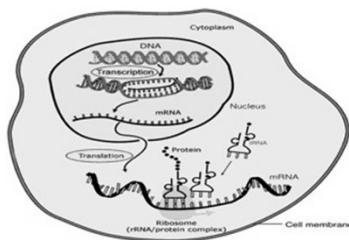


RNA

- ❖ RNA contains the bases A, G, C, and U instead of A, G, C, and T.
- ❖ RNA exists as a single strand instead of a double strand. There are three primary types of RNA. They are
 1. Messenger RNA (m-RNA)
 2. Ribosomal RNA (r-RNA)
 3. Transfer RNA (t-RNA)
- ❖ The different kinds of RNA perform different functions. RNA makes protein synthesis possible.
- ❖ The three RNA make it possible for the encoded information carried by the DNA to be put to use in the synthesis of proteins.

RNA

- ❖ Messenger RNA (mRNA) are of variable size, depending on the protein to be manufactured, and contain the information that specifies which protein should be made. This message is a sequence of RNA nucleotides that is complementary to the template strand of DNA.
- ❖ Ribosomal RNAs (rRNA) are relatively long RNA strands (hundreds or thousands of nucleotide residues) that combine with proteins to form ribosomes, the multisubunit complexes in which protein synthesis takes place.
- ❖ Transfer RNA (tRNA) are the smallest of the three types (73-93 nucleotide residues), and they carry the correct amino acid to the site of protein synthesis.



Difference between RNA & DNA

S.No.	RNA	DNA
1)	Single stranded mainly except when self complementary sequences are there it forms a double stranded structure (Hair pin structure)	Double stranded (Except for certain viral DNA s which are single stranded)
2)	Ribose is the main sugar	The sugar moiety is deoxy ribose
3)	Pyrimidine components differ. Thymine is never found (Except tRNA)	Thymine is always there but uracil is never found
4)	Being single stranded structure- It does not follow Chargaff's rule	It does follow Chargaff's rule. The total purine content in a double stranded DNA is always equal to pyrimidine content.

THANK YOU